WORKED SOLUTION

STEP I 1999, Question 11

Mechanics • Gravitational Circular Motion

STEP I 1999, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q11

This question has a full guided treatment in *Beyond STEP*, Section 10.8, *Three Stars in a Triangle*: https://beyondstep.keithmcnulty.org/ch10/three-stars.html

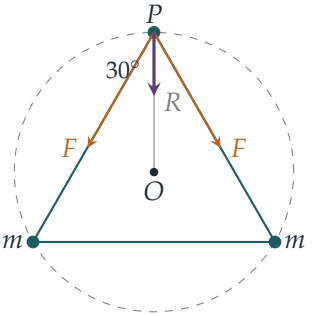
The key idea

Guess the motion and test it. If the triangle turns rigidly about its centre, each star moves in a circle, so it needs a net force of exactly $m\omega^2 R$ pointing to the centre. By symmetry, the gravitational pull on each star from the others points straight at the centre. So all we have to do is match its size to $m\omega^2 R$ and read off ω. A star placed at the centre adds one more inward pull, of a known size.

Solution

We model the stars as particles that move only under their mutual attractions. Let O be the centre of mass. For three equal masses this is the centroid of the triangle, and its distance from each vertex is

$$R = \tfrac{2}{3} \times \tfrac{\sqrt{3}}{2} a = \frac{a}{\sqrt{3}}.$$



Three stars

Take the star P at the top vertex. Each of the other two stars pulls it with a force of size

$$F = \frac{\gamma m^2}{a^2},$$

directed along a side of the triangle. Each side through P makes an angle of 30° with PO, because PO bisects the 60° angle at P. Resolving along and perpendicular to PO:

- perpendicular to PO, the two components are equal and opposite, so they cancel;
- along PO, the resultant is $2F\cos 30^\circ = \sqrt{3}\,\dfrac{\gamma m^2}{a^2}$, towards O.

The other two stars are in the same situation, by symmetry.

Now suppose the triangle rotates rigidly about O with angular velocity ω. Each star then moves in a circle of radius R with constant speed $R\omega$, so it needs an acceleration $R\omega^2$ towards O. Newton's second law towards O gives

$$m\,\frac{a}{\sqrt{3}}\,\omega^2 = \sqrt{3}\,\frac{\gamma m^2}{a^2} \quad\Longrightarrow\quad \omega^2 = \frac{3\gamma m}{a^3}.$$

Ask yourself Why does this show that the motion is *possible*, and not just that the forces balance at one instant?

Start the stars at the vertices, with velocities of size $R\omega$ perpendicular to OP, all turning the same way. While the configuration stays an equilateral triangle of side a centred at O, the force on each star has exactly the size and direction that circular motion with angular velocity ω requires. So the rigidly rotating triangle satisfies the equation of motion of every star at every moment. It is therefore a genuine motion of the system: it is the solution that follows from those initial conditions. The total momentum is zero, so O stays fixed, as it should. Hence each star can revolve about the centre of mass with angular velocity $(3\gamma m/a^3)^{1/2}$.

Adding a fourth star at the centre

Place a star of mass λm at O. The centre of mass is still O. The three outer stars pull the central star with forces of equal size at 120° to one another, so the resultant is zero. The central star therefore stays at rest at O, provided that the outer three keep their symmetric arrangement.

Each outer star now has an extra pull towards O, of size

$$\frac{\gamma(\lambda m)m}{R^2} = \frac{3\gamma\lambda m^2}{a^2},$$

since $R^2 = a^2/3$. This extra force acts along PO, so it simply adds to the inward

resultant. The equation of motion towards O becomes

$$m\,\frac{a}{\sqrt{3}}\,\omega^2 = \sqrt{3}\,\frac{\gamma m^2}{a^2} + 3\lambda\,\frac{\gamma m^2}{a^2}$$

$$\omega^2 = \frac{3\gamma m}{a^3} + \frac{3\sqrt{3}\,\lambda\gamma m}{a^3} = \frac{3\gamma m}{a^3}\left(1+\sqrt{3}\,\lambda\right).$$

The same argument as before shows that this rigid rotation is a genuine motion. So, with the fourth star at rest at the centre, the outer three can revolve about it with angular velocity

$$\omega = \left(\frac{3\gamma m\,(1+\sqrt{3}\,\lambda)}{a^3}\right)^{1/2}.$$

Checks

- Setting $\lambda = 0$ gives back the first answer, as it must.
- Dimensions: $\gamma m/r^2$ is an acceleration, so $\gamma m/a^3$ has units of time^{-2}, which is right for ω^2.
- If λ is very large, the outer stars barely feel one another. They then orbit the central mass alone, with $\omega^2 \approx \gamma\lambda m/R^3 = 3\sqrt{3}\,\gamma\lambda m/a^3$, which matches the dominant term.
- We integrated the full equations of motion numerically (with $\gamma = m = a = 1$). Starting from the predicted circular velocities, the stars returned to their starting positions after three revolutions, with relative error below 10^{-7}, for $\lambda = 0$, 1 and 2.5. For example, $\lambda = 1$ gives $\omega^2 = 3(1+\sqrt{3}) \approx 8.196$.

Common slips

- Using a instead of $R = a/\sqrt{3}$ as the radius of the circle, or as the distance to the central star.
- Adding the two pulls of size F as $2F$, forgetting that they are at 30° to PO.
- Forgetting to say why the central star stays put: the three pulls on it cancel.

Beyond STEP

This is a special case of Lagrange's equilateral solution of the three-body problem, which works for *any* three masses at the vertices of a rotating equilateral triangle. The same idea gives the Lagrange points L_4 and L_5, where the Trojan asteroids share Jupiter's orbit. Whether such configurations are *stable* is a deeper question, answered by linearising the equations of motion about the rotating solution. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats central forces and orbits carefully at this level.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.